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 Studierichting: Informatica
 Oefeningenreeks 3G nr. 13.2

$$f(x) = \sin 3x$$

$$\text{Periode} = \frac{2\pi}{3}$$

$$f^{-1}(0) = \left\{0, \frac{\pi}{3}, \frac{2\pi}{3}\right\}$$

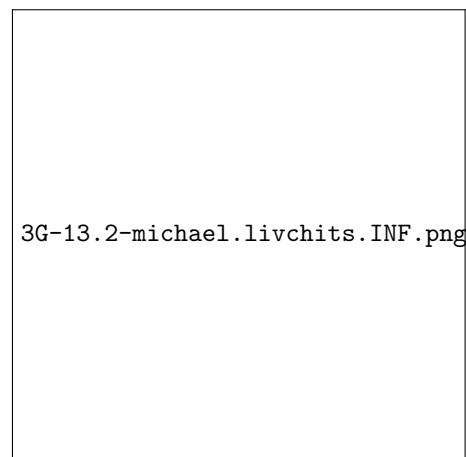
$$f'(x) = 3 \cos 3x$$

$$(f')^{-1}(0) = \left\{\frac{\pi}{6}, \frac{\pi}{2}\right\}$$

$$f''(x) = -9 \sin 3x$$

$$(f'')^{-1}(0) = \left\{0, \frac{\pi}{3}, \frac{2\pi}{3}\right\}$$

x	0		$\frac{\pi}{6}$		$\frac{\pi}{3}$		$\frac{\pi}{2}$		$\frac{2\pi}{3}$
$f(x)$	0	+	+	+	0	−	−	−	0
$f'(x)$	+	+	0	−	−	−	0	+	+
$f''(x)$	0	−	−	−	0	+	+	+	0
$f(x)$	$\nearrow$ $B(0)$	$\nearrow$ $\smile$	$M(1)$ $\smile$	$\searrow$ $\smile$	$\searrow$ $B(0)$	$\searrow$ $\smile$	$m(-1)$ $\smile$	$\nearrow$ $\smile$	$\nearrow$ $B(0)$



References